

Power BI Syllabus

DAY 1

1. What is data analytics and business analytics
2. Business Analyst v/s Data Analyst
3. What is PowerBI
4. Why do we need Power BI
5. Tableau V/s Power BI
6. How a business problem is resolved [methodology]+ work of Power BI in it.
7. Power BI Components
 - Power BI Desktop
 - Power BI Services
 - Power BI Mobile
8. Installation of Power BI

DAY 2

1. Entire Power BI flow (Data collection → Power BI Desktop → Power BI Services)
 - What are Servers
 - How data is Stored
 - ETL process
 - OLAP/OLTP/Data Warehouses, Data Lakes, Data Marts etc.
 - Data Cleaning
 - Data Transformation.
 - Data Visualization.
 - Power Query
 - DAX
 - Measures Etc.
 - Publishing of dashboard, Self Service BI, Refresh Reports, Sharing of reports.
2. Power BI user Interface.
3. Data types available, their Identification, and Changing Data Types.

DAY 3 AND DAY 4

1. Editing working area
2. Creating and editing visuals.
3. Elements such as cards, slicers, KPI, gauge, etc.
4. Bookmarks, hyperlinks, and page navigation.
5. Format Painter.
6. Area chart [Lasso setting], Area bubble chart, Area filled chart.
7. Integration of AI to create charts.
8. Tables, matrix creations, and differences between them.
9. Conditional Formatting.

DAY 5

1. Tooltips.
2. Filters.
3. Drill up, Drill down, Drill Through.
4. Data Modelling (Data Relationships).
 - Cardinality.
 - Relationships.
 - Cross Filter Direction.

DAY 6 AND DAY 7

1. Data Modelling Continuation.
2. Filters v/s Slicers [performance].
3. Introduction to DAX
 - Basic Dax Queries.
 - If, switch Statements
 - DAX and logic building practice.

DAY 8 AND DAY 9

1. Consolidating Data (Grouping, Bins)
2. Measures (Continuation of DAX)
 - Questions and logic.
 - Difference between measures and columns.
 - Types of measures.

DAY 10

1. Filter Context.
2. Queries and Logic building in Filter Context (DAX)

DAY 11

1. Parameterized Charts Continuation.
2. measure Parameterized Charts + DAX.
3. Tree Map.
4. Scatter Plot.

DAY 12

1. Quick Measures.
2. Introduction to Power Query.
3. Problems and issues with data and continuation of Power Query
4. Data Transformation using Power Query Editor.

DAY 13 AND DAY 14

1. Data transformation Continuation.
2. M- language.
3. Merge Queries.
4. Append Queries.
5. Connect of data from Servers (SQL servers)
6. Query Folding.
7. Python Integration.

DAY 15

1. Pivot and Unpivot tables.
2. Granularity and Group by.
3. Row Level Security in Power BI.
4. Power BI Services Introduction.
5. DAX and measures using AI.

DAY 16

1. Power BI Services.
 - Publishing of Services.
 - Assignment of roles.
 - WorkSpaces.
 - Self Service BI.
 - Managed Access.
2. Power BI pricing.

DAY 17

1. Power BI roles.

2. Self-Service Capabilities.
3. Reports (In Power BI Services).
4. Dashboard (In Power BI Services).
5. Reports V/S Dashboards.

DAY 18

1. Power BI Architecture.
2. Gateways.
3. Refresh Time.
4. Embedded links.

DAY 19

1. Power BI Interview Questions.

DAY 20

1. Gen AI Enabled Power BI
 - AI charts.
 - Analysis Using AI to Drive Insights from Created Charts.
 - Formatting background using AI.
 - Formatting charts using AI.

DAY 21,22,23, 24, 25

1. Power BI projects + Microsoft Power BI PL 300 certification Guidance.

PROJECTS [Tentative]

1- Global SuperStore Dataset(class)

https://docs.google.com/spreadsheets/d/1HKlafg7vmhzW7p_p040qFXLOdcBzj1HCseNjkmOeZ8/edit?usp=sharing

2- Russia Ukraine War [HUMANITARIAN ISSUES]

<https://www.kaggle.com/datasets/piterfm/2022-ukraine-russian-war>

3- Amazon Sales Analysis (report)

<https://www.kaggle.com/datasets/thedevastator/unlock-profits-with-e-commerce-sales-data?select=Amazon+Sale+Report.csv>

4- Superstore Sales Data.

<https://www.kaggle.com/datasets/rohitsahoo/sales-forecasting>

5- Transport and shipment. [LOGISTICS & SUPPLY CHAIN]

<https://www.kaggle.com/datasets/anisseezzebdi/supply-chain-logistics-problem>

6- Football world cup Analysis. [SPORTS]

https://www.kaggle.com/datasets/piterfm/fifa-football-world-cup?select=matches_1930_2022.csv

Or

<https://www.kaggle.com/datasets/dle9origephit/fifa-world-cup-2022-complete-dataset>

7- IBM , human resource attrition. [HR]

<https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset>

8- LinkedIn Job posting [HR]

<https://www.kaggle.com/datasets/arshkon/linkedin-job-postings?select=mappings>

9- India Census 2011. [NATIONAL]

<https://www.kaggle.com/datasets/danofer/india-census>

10- Nobel Prize Winners. [WINNERS AND AWARDS]

<https://www.kaggle.com/datasets/joebeachcapital/nobel-prize>